

CAO BANG, Viet Nam

WMO: 488080

Lat: 22.6636N

Lon: 106.2510E

Elev: 243

StdP: 98.44

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.2	7.8	2.1	4.5	12.0	3.8	5.1	11.4	6.8	20.0	5.7	18.9	0.7	180	0.282

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.2	35.0	27.1	34.0	26.8	33.1	26.6	28.2	33.2	27.7	32.4	27.2	31.7	2.3	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.9	23.3	31.5	26.5	22.6	30.6	26.1	22.1	29.9	93.1	33.4	90.6	32.7	88.3	31.9	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.9	4.8	4.0	3.9	37.4	1.7	0.9	2.7	38.1	1.7	38.7	0.8	39.2	-0.4	39.9		
			3.6	29.5	1.5	0.7	2.5	30.0	1.6	30.4	0.8	30.8	-0.3	31.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.7	14.6	16.7	19.6	23.9	26.5	28.0	28.2	27.9	26.7	24.1	20.2	16.0	
	DBStd	5.65	3.72	4.38	3.84	3.49	2.63	1.73	1.53	1.51	1.91	2.39	2.96	3.19	
	HDD10.0	5	3	1	0	0	0	0	0	0	0	0	0	1	
	HDD18.3	343	127	79	31	3	0	0	0	0	0	1	14	88	
	CDD10.0	4654	146	189	298	417	512	541	564	554	502	437	307	187	
	CDD18.3	1951	12	33	71	171	253	291	306	296	252	179	71	17	
	CDH23.3	16806	40	181	371	1270	2387	3023	3124	2789	2107	1107	342	64	
CDH26.7	6136	3	44	105	430	932	1169	1203	1052	782	337	72	7		
Wind	WSAvg	1.4	1.3	1.5	1.7	1.8	1.5	1.4	1.3	1.2	1.1	1.1	1.2	1.2	
Precipitation	PrecAvg	1493	36	29	54	81	197	279	273	254	129	81	46	34	
	PrecMax	1782	125	78	217	184	314	459	544	357	258	245	137	103	
	PrecMin	1110	5	3	4	17	104	120	125	51	54	17	3	7	
	PrecStd	188	27	23	53	41	55	95	80	82	61	49	35	27	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.9	30.5	32.4	35.8	36.5	36.0	35.9	35.3	34.4	32.7	30.4	27.6	
		MCWB	20.2	21.5	23.1	24.7	26.9	27.5	27.8	27.3	26.4	24.7	23.6	21.4	
	2%	DB	24.2	28.0	29.5	32.8	34.3	34.3	34.4	34.1	33.3	31.2	28.6	25.1	
		MCWB	19.1	21.3	23.1	25.2	26.9	27.4	27.4	27.0	26.0	24.0	22.5	19.8	
	5%	DB	22.3	25.6	27.3	30.9	32.8	33.1	33.3	33.0	32.2	30.0	27.0	23.0	
		MCWB	17.9	20.3	22.1	24.6	26.1	27.0	27.1	26.8	25.6	23.4	21.3	18.3	
	10%	DB	20.3	23.2	25.1	29.2	31.3	32.0	32.0	31.8	31.0	28.7	25.2	21.1	
		MCWB	17.1	19.3	21.1	23.9	25.3	26.7	26.8	26.5	24.9	23.1	20.5	17.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.8	22.6	24.4	26.6	28.4	28.8	28.8	28.6	27.7	26.2	24.7	22.3
			MCDB	25.2	28.0	29.9	32.2	34.2	33.8	33.9	33.4	32.3	30.5	28.8	26.4
		2%	WB	19.6	21.6	23.4	25.7	27.4	28.1	28.1	27.9	27.0	25.2	23.7	20.7
			MCDB	23.3	26.6	28.1	30.9	32.9	32.9	33.0	32.7	31.5	29.3	27.2	23.9
5%		WB	18.7	20.7	22.6	25.0	26.7	27.5	27.6	27.2	26.4	24.5	22.7	19.4	
		MCDB	21.7	24.9	26.4	29.8	31.5	32.0	32.1	31.6	30.6	28.2	25.5	22.2	
10%		WB	17.6	19.8	21.7	24.3	26.0	27.0	27.0	26.7	25.9	23.9	21.7	18.0	
		MCDB	19.9	22.9	24.7	28.4	30.1	31.1	31.1	30.7	29.6	27.2	24.0	20.3	
Mean Daily Temperature Range		5% DB	MDBR	6.5	6.9	6.6	7.2	7.7	7.2	7.2	7.6	8.2	8.3	8.5	8.0
			MCDBR	10.3	11.3	10.1	10.1	10.2	9.2	9.4	9.7	10.0	10.3	10.9	10.7
	MCWBR		5.0	5.0	4.3	3.8	3.7	3.3	3.3	3.5	3.2	3.7	5.0	5.4	
	5% WB	MCDBR	8.7	10.1	9.1	9.0	8.8	8.2	8.4	8.8	8.6	8.5	8.5	9.0	
		MCWBR	4.5	4.8	4.2	3.8	3.7	3.5	3.5	3.6	3.3	3.5	4.5	5.0	
Clear-Sky Solar Irradiance	taub	0.579	0.601	0.772	0.730	0.550	0.535	0.509	0.569	0.601	0.686	0.580	0.606		
	taud	1.744	1.737	1.451	1.573	2.029	2.090	2.152	1.963	1.850	1.621	1.794	1.699		
	Ebn at Noon	676	698	605	646	770	775	796	751	717	624	671	631		
	Edh at Noon	209	223	310	278	175	164	154	186	205	247	197	212		
All-Sky Solar Radiation	RadAvg	2.03	2.58	2.91	3.75	4.38	4.47	4.71	4.67	4.37	3.47	3.03	2.37		
	RadStd	0.48	0.51	0.34	0.45	0.49	0.48	0.32	0.31	0.30	0.30	0.44	0.38		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only			+0.32	N/A	N/A	+0.71	+0.60	+0.51	N/A	N/A	N/A
Regional (6 neighbors)			N/A	N/A	N/A	+0.57	N/A	N/A	N/A	N/A	+80

Nomenclature: See separate page